

VOLUME 2

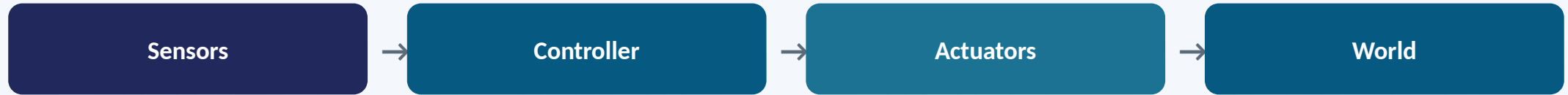
Robot Engineering & Autonomy

From robot fundamentals and embedded systems to perception, motion, ROS 2, navigation, and a complete budget vision robot capstone.

Chapter 1 • 2 • 3 • 4 • 5 • 6 • 7 • 8 • 9 • 10

Robot Fundamentals & Design Principles

A robot senses, decides, and acts. This chapter covers anatomy, degrees of freedom, kinematics, categories, and safety.



ANATOMY

Base • Links • Joints • Actuators • Sensors • End Effector

DOF & KINEMATICS

Independent movements (up to 6 DOF); Forward Kinematics maps angles → position, Inverse Kinematics maps position → angles

CATEGORIES

Industrial (fixed, precise) • Service • Mobile • Humanoid — each trades precision for adaptability

SAFETY FIRST

Disconnect power before wiring, verify battery polarity, use an emergency stop, test at low speed

Embedded Systems & ESP32 Mastery

A dedicated computer designed for a specific task. The ESP32 pairs a dual-core CPU with Wi-Fi and Bluetooth for robotics and IoT.

MICROCONTROLLER (ESP32)

Built-in RAM/Flash · Low power · Real-time control

MICROPROCESSOR (Raspberry Pi CPU)

Needs external RAM · Higher power · General computing

ESP32 CORE FEATURES

CPU

GPIO

Wi-Fi

Bluetooth

RAM

Flash

COMMUNICATION PROTOCOLS

UART

Serial communication

I²C

Sensors and OLED displays

SPI

Fast displays and memory

CAN

Automotive and industrial systems

FreeRTOS & power: run sensor reading, motor control and comms as separate tasks; use deep sleep; never power motors directly from GPIO.

Sensors, Cameras & Robot Perception

Robots observe their environment through sensors and cameras, interpret the data, and make decisions — improving navigation and safety.

ANALOG SENSORS

Continuous voltage, e.g. LDR, potentiometer

DIGITAL SENSORS

Discrete, stable signal, e.g. ultrasonic, IMU, GPS

SENSOR & CAMERA TYPES

Ultrasonic

Reflected sound — obstacle avoidance

IR / ToF

Short-range or precise light-based ranging

IMU

Accelerometer + gyroscope — balance & orientation

RGB / Depth / Stereo

Vision — object detection, 3D distance

Engineer tips: match sensors to range/accuracy needs, mount firmly to reduce vibration, calibrate before use, and fuse multiple sensors for reliability.

Motors, Drivers & Motion Control

Motors are the most common actuator in robotics. Drivers safely deliver power; control algorithms produce smooth, accurate motion.

MOTOR	ADVANTAGE	APPLICATIONS
DC Motor	Simple, inexpensive	Robot cars, conveyors
Servo Motor	Precise angle control	Robot arms, pan-tilt
Stepper Motor	High positioning accuracy	3D printers, CNC
BLDC Motor	Efficient, high speed	Drones, AGVs, e-bikes

Drive systems: Differential Drive (two wheels) · Ackermann Steering (cars) · Omni/Mecanum Wheels (sideways movement)

Batteries: choose by voltage, capacity (Ah), and discharge current — Li-ion/LiPo are common. Never power motors directly from GPIO pins.

ROS 2 Fundamentals

An open-source robotics framework providing communication, hardware abstraction, visualization, and development tools — not an OS itself.

Nodes

Independent programs

Topics

Publish / subscribe communication

Services

Request-response interactions

Actions

Long-running tasks with feedback

Messages

Structured data between nodes

TOOLS & INFRASTRUCTURE

RViz

Visualizes robot models, sensor data, maps

Gazebo

Simulate robots in virtual environments

TF

Tracks coordinate frames consistently

Why ROS 2: modular design, real-time support, DDS-based networking, cross-platform, and a large ecosystem of reusable packages.

Computer Vision for Robotics

Computer vision transforms pixels into useful information — object identity, location, orientation, and movement.

Camera

Preprocessing

AI Model

Decision

Robot Action

Image Processing

Cleaning images

Object Detection

Find objects (e.g. YOLO)

Tracking

Follow motion across frames

Segmentation

Separate objects from background

Pose Estimation

Estimate orientation

OCR

Read text in images

Engineer tips: use good lighting, calibrate cameras, start with OpenCV before advanced AI models, and optimize image size for real-time performance.

Robot Navigation & SLAM

Navigation combines mapping, localization, perception and planning so a robot can move safely from one place to another.

Perception

Localization

Mapping

Planning

Motion Control

SLAM (Simultaneous Localization And Mapping)

LiDAR SLAM

Visual SLAM

RGB-D SLAM

Graph-based SLAM

Odometry

Simple and fast

Short-distance estimation

A*

Optimal path

Known maps

RRT

Exploration

Complex environments

Engineer tips: fuse multiple sensors, calibrate encoders/IMUs regularly, and always test navigation algorithms in simulation first.

Building Your First Intelligent Robot

Integrating chassis, controller, sensors, motors, and camera into a small mobile robot that senses, moves, and responds to commands.

Mechanical Assembly

Robot chassis completed

Electrical Wiring

All components connected safely

Basic Programming

Motors and sensors working

Camera Integration

Robot captures images

Testing

Reliable autonomous behaviour

Optimization

Improved speed and accuracy

REQUIRED COMPONENTS

ESP32 dev board · robot chassis · 2 DC geared motors · motor driver · battery pack · ultrasonic sensor · optional camera · jumper wires · breadboard · switches

Engineer tips: label every wire, test each module independently, secure loose cables, and back up your code with Git.

Simulation, Testing & Engineering Practices

Simulation lets engineers test robots safely before deploying to real hardware — reducing cost and risk.

PLATFORM	STRENGTH	TYPICAL USE
Gazebo	ROS integration	Mobile robots
Webots	Easy learning	Education & research
Isaac Sim	High-fidelity physics	AI & industrial robotics
Unity / Unreal	Rich / photorealistic graphics	Visualization

Testing hierarchy: unit testing → subsystem testing → integration testing → field testing.

Key metrics: accuracy, precision, latency, frame rate, battery life, navigation success rate, reliability.

Documentation: wiring diagrams, software architecture, version history, experiment logs, user manuals.

Capstone: Budget Vision Robot

Integrating hardware and software into a low-cost intelligent robot capable of perception, navigation, and basic autonomous behavior.

Sense**Process****Decide****Act****Repeat**

BILL OF MATERIALS

ESP32

Main controller

Ultrasonic Sensor

Obstacle detection

Battery

Power

Motor Driver

Controls motors

Camera

Computer vision

Robot Chassis

Mechanical structure

CAPSTONE CHECKLIST

☐ Power verified ☐ Motors tested ☐ Sensors calibrated ☐ Camera operational ☐ Object detection demonstrated ☐ Documentation completed

From Fundamentals to a Working Vision Robot

Volume 2 carried you from robot anatomy and embedded systems through perception, motion control, ROS 2, computer vision, navigation, and a complete budget vision robot build.

Next: advanced Vision-Language-Action models in Volume 3.